

XRPL Liquidity Pool (AMM)

Implementation Plan

DeGears Backend — Technical Documentation

Overview

XRPL has native AMM (Automated Market Maker) support via the XLS-30 amendment. This means liquidity pools are built into the ledger itself — no smart contracts needed. We pair the memecoin with XRP so anyone can trade it.

How XRPL AMM Works

1. Creator creates MemeCoin (already done)
2. Creator creates AMM Pool (MemeCoin / XRP pair)
3. Creator deposits initial liquidity (e.g., 500,000 OTG + 1,000 XRP)
4. XRPL mints LP Tokens to the creator (proof of liquidity share)
5. Anyone can now swap XRP <-> MemeCoin via the AMM
6. Liquidity providers can deposit/withdraw at any time

XRPL AMM Transaction Types

Transaction	Purpose
AMMCreate	Create a new AMM pool for a token pair
AMMDeposit	Add liquidity to an existing pool
AMMWithdraw	Remove liquidity from a pool
AMMBid	Bid on the auction slot (discounted trading fee)
AMMVote	Vote on the pool's trading fee
amm_info	Query pool info (reserves, LP token supply, fee)

API Design (3-Step Flow)

Step 1: POST /api/v1/memecoins/:id/create-pool

Builds an AMMCreate transaction for the creator to sign via Xaman QR.

Request Payload:

```
{
  "walletAddress": "rCreator...",
  "tokenAmount": "500000",
  "xrpAmount": "1000"
}
```

AMMCreate Transaction:

```
{
  "TransactionType": "AMMCreate",
  "Account": "rCreator...",
  "Amount": {
    "currency": "OTG",
    "issuer": "rIssuer...",
    "value": "500000"
  },
  "Amount2": "1000000000",
  "TradingFee": 500
}
```

- Amount = token amount to deposit as initial liquidity
- Amount2 = XRP in drops (1000 XRP = 1,000,000,000 drops)
- TradingFee = fee in basis points (500 = 0.5%)
- Creator signs via Xaman, then XRPL creates the pool and mints LP tokens

Step 2: POST /api/v1/memecoins/:id/confirm-pool

After creator signs, verify the AMMCreate tx on XRPL and save pool details.

Request Payload:

```
{
  "ammCreateTxHash": "ABC123..."
}
```

Backend Actions:

1. Verify tx on XRPL (command: tx)
2. Fetch pool info (command: amm_info)
3. Save pool details to DB (new LiquidityPool model)
4. Update memecoin status to "pooled"

Step 3: POST /api/v1/memecoins/:id/add-liquidity (Optional)

Lets any user add liquidity to an existing pool.

AMMDeposit Transaction:

```
{
  "TransactionType": "AMMDeposit",
  "Account": "rUser...",
  "Asset": {
    "currency": "OTG",
    "issuer": "rIssuer..."
  },
  "Asset2": {
    "currency": "XRP"
  },
  "Amount": {
    "currency": "OTG",
    "issuer": "rIssuer...",
    "value": "10000"
  },
  "Amount2": "200000000",
  "Flags": 1048576
}
```

Additional Endpoints

Method	Endpoint	Description
GET	/api/v1/memecoins/:id/pool-info	Get AMM pool info (reserves, price, LP supply, fee)
POST	/api/v1/memecoins/:id/remove-liquidity	Build AMMWithdraw tx for QR signing
POST	/api/v1/memecoins/:id/swap	Build a Payment with DeliverMin for token swap
GET	/api/v1/memecoins/pools	List all active liquidity pools

Database Model: LiquidityPool

Field	Type	Description
id	UUID	Primary key
memeCoinId	UUID (FK)	References MemeCoins.id
ammAccountId	STRING	XRPL AMM account address
tokenCurrency	STRING	Currency hex code
tokenIssuer	STRING	Issuer address
pairedWith	STRING	Default: XRP
tradingFee	INTEGER	Basis points (e.g., 500 = 0.5%)
initialTokenAmount	DECIMAL	Initial token deposit
initialXrpAmount	DECIMAL	Initial XRP deposit
creatorWalletAddress	STRING	Pool creator wallet
ammCreateTxHash	STRING	AMMCreate transaction hash
status	ENUM	active / empty / removed
lpTokenCurrency	STRING	LP token currency code
metadata	JSON	Additional metadata

Trading Fee Recommendations

Fee	Use Case
0 (0%)	Maximum volume, no LP revenue
100 (0.1%)	Low fee, high volume
500 (0.5%)	Recommended for memecoins &
1000 (1%)	Higher fee, less volume
	Max allowed by XRPL: 1000 (1%)

How Swaps Work

Once the AMM pool exists, anyone can swap using a standard XRPL Payment with path-finding. XRPL auto-routes through the AMM. The frontend can build swap transactions directly — the backend just needs to provide pool info and token details.

```
{
  "TransactionType": "Payment",
  "Account": "rBuyer...",
}
```

```
"Destination": "rBuyer...",  
"Amount": {  
  "currency": "OTG",  
  "issuer": "rIssuer...",  
  "value": "1000"  
},  
"SendMax": "500000000",  
"Flags": 131072  
}
```

Implementation Priority

Phase	What	Endpoints
Phase 1	Create pool + confirm	create-pool, confirm-pool, pool-info
Phase 2	Add/remove liquidity	add-liquidity, remove-liquidity
Phase 3	Swap helper + pool listing	swap, pools

Key Considerations

1. Account : XRPL creates a special AMM account automatically — you don't manage it
2. LP Tokens: Minted by XRPL, represent liquidity share, can be traded
3. One Pool Per Pair : XRPL

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